



DS*ci* | Study
& Research

Data Science Journey

Mathematics

Domain
Knowledge

Computer
Science

5 years 11 months have passed

since the start of the journey on 3 January 2020.

Abstract

Data science is the study of data to extract meaningful insights for business. It is a multi-disciplinary approach that combines principles and practices from the fields of mathematics, statistics, artificial intelligence, and computer engineering to analyze large amounts of data.

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Teil I

Mathematics

Kapitel 1

Markov Chains

1 The Law of Large Numbers (LLN)

The **Law of Large Numbers (LLN)** is a fundamental theorem in probability theory that describes how averages behave over many repetitions of a random experiment. It ensures that the average of repeated independent observations converges to the expected (theoretical) mean.

1.1 Key Symbols

Symbol	Meaning
X_i	Random variable representing the i -th trial
$\mu = \mathbb{E}[X_i]$	Expected (mean) value
$\bar{X}_n$	Sample mean after n observations
$\xrightarrow{P}$	Convergence in probability
$\xrightarrow{\text{a.s.}}$	Almost sure convergence

1.2 Setup

Let

$$\{X_1, X_2, \dots\}$$

be a sequence of independent and identically distributed (i.i.d.) random variables with finite expected value

$$\mathbb{E}[X_i] = \mu.$$

Define the sample mean:

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

1.3 Weak Law of Large Numbers (WLLN)

The sample mean converges to the expected value *in probability*:

$$\bar{X}_n \xrightarrow{P} \mu \quad \text{as } n \rightarrow \infty,$$

that is, for every $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

1.4 Strong Law of Large Numbers (SLLN)

The sample mean converges to μ *almost surely*:

$$\overline{X}_n \xrightarrow{\text{a.s.}} \mu \quad \text{as } n \rightarrow \infty,$$

meaning

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \overline{X}_n = \mu\right) = 1.$$

1.5 Interpretation

The LLN justifies the use of empirical averages to estimate theoretical expectations. As the number of samples grows, randomness averages out, and the observed mean stabilizes around the true mean.

2 Markov Chains and Dependence

2.1 Key Symbols

Symbol	Meaning
Markov property	$\mathbb{P}(X_{n+1} X_n, \dots) = \mathbb{P}(X_{n+1} X_n)$ (local forgetting)
Ergodicity	Global forgetting; convergence to a unique π
π	Stationary distribution ($\pi P = \pi$)
$S = \{s_1, \dots, s_k\}$	Finite set of possible states
X_n	State of the process at step n
$\mathbb{P}(X_{n+1} = s_j X_n = s_i)$	Transition probability

2.2 Definition

A **Markov chain** is a sequence of random variables

$$X_1, X_2, X_3, \dots$$

taking values in a state space

$$S = \{s_1, s_2, \dots, s_k\},$$

and satisfying the *Markov property*:

$$\mathbb{P}(X_{n+1} = s_j | X_n = s_i, X_{n-1}, \dots, X_1) = \mathbb{P}(X_{n+1} = s_j | X_n = s_i), \quad \forall i, j.$$

2.3 Interpretation

The *Markov property* is a local forgetting statement: conditional on the present state, the future is independent of the past. By itself, this property does *not* imply convergence of empirical averages.

However, if a Markov chain satisfies additional regularity conditions that force *global forgetting* (ergodicity)—for example, on a finite state space: irreducibility, aperiodicity, and positive

recurrence—then the chain admits a unique stationary distribution π , the law of X_n converges to π regardless of the start, and a Law of Large Numbers holds for functions of the chain.

In General: Markov showed that independence is *not necessary* for the Law of Large Numbers to hold. Even when consecutive events are dependent—as in a Markov chain—the average value still converges to a stable limit if the dependence is sufficiently regular (e.g., the chain is stationary or ergodic). Thus, the LLN can be “caused by dependent events.”

Case	Independence	Expectation	LLN Holds?
i.i.d.	Yes	Constant (μ)	Yes
Stationary Markov Chain	No	Constant (μ)	Yes
Ergodic Markov Chain	No	Converges to μ	Yes

In particular, independence of the X_i is not necessary for an LLN: what is needed is sufficient mixing/forgetting of the initial state, as guaranteed by ergodicity.

3 Probability Distributions on Finite and Infinite State Spaces

Let S be the state space of a Markov chain. A *probability distribution* on S assigns to each state $i \in S$ a nonnegative number $\pi(i)$ representing the probability that the chain is in state i , with the probabilities summing to one.

3.1 Finite State Space.

If the state space is finite,

$$S = \{s_1, s_2, \dots, s_k\},$$

then a probability distribution on S is a vector

$$\pi = (\pi(s_1), \dots, \pi(s_k)),$$

satisfying

$$\pi(s_j) \geq 0, \quad \sum_{j=1}^k \pi(s_j) = 1.$$

In this case π is simply a finite probability vector.

The full probability distribution at time n . The collection of all such probabilities,

$$\pi^{(n)} = (\mathbb{P}(X_n = s_1), \mathbb{P}(X_n = s_2), \dots),$$

is the probability distribution of X_n . Thus,

$$\pi^{(n)}(j) = \mathbb{P}(X_n = s_j).$$

3.2 Countably Infinite State Space.

If the state space is countably infinite,

$$S = \{s_1, s_2, s_3, \dots\},$$

then a probability distribution is an infinite sequence

$$\boldsymbol{\pi} = (\pi(s_1), \pi(s_2), \dots),$$

satisfying

$$\pi(s_j) \geq 0 \quad \text{for all } j, \quad \sum_{j=1}^{\infty} \pi(s_j) = 1.$$

The normalization condition is now an infinite series, which must converge to 1.

3.3 Interpretation

In both cases, $\pi(i)$ gives the probability that $X_n = i$ when the chain is in distribution $\boldsymbol{\pi}$. For finite state spaces the distribution is a finite probability vector, while for infinite state spaces it is an infinite but summable probability sequence.

4 Evolution and Convergence

4.1 Key Symbols

Symbol	Meaning
P_{ij}	One-step transition probability
P	Transition matrix
P^n	n -step transition probabilities
Stationary $\boldsymbol{\pi}$	Limiting equilibrium distribution
$\boldsymbol{\pi}^{(n)}$	Distribution at step n
$\boldsymbol{\pi}^{(0)}$	Initial distribution vector
$\boldsymbol{\pi}$	Stationary distribution
μ	Expected value under stationarity

4.2 Key Symbols - Transition Matrix

Symbol	Meaning
$p_{VV}, p_{VC}, p_{CV}, p_{CC}$	Transition probabilities
π_V, π_C	Stationary probabilities
$\boldsymbol{\pi}^{(0)}$	Initial distribution
P^n	n -step transition matrix

4.3 Transition Matrix

The one-step transition probabilities are collected into a matrix:

$$P = \begin{bmatrix} P_{11} & P_{12} & \cdots & P_{1k} \\ P_{21} & P_{22} & \cdots & P_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ P_{k1} & P_{k2} & \cdots & P_{kk} \end{bmatrix}, \quad \sum_{j=1}^k P_{ij} = 1.$$

5 The Idea of Forgetting - Stationary and non-stationary Distributions

5.1 Key Symbols

Symbol	Meaning
P_{ij}	Probability of moving from state s_i to s_j
π_j	Long-run (stationary) probability of state s_j
π	Stationary distribution vector
Ergodic chain	Chain that “forgets” its start and converges to π

5.2 Overview

Case	Independence	Expectations	LLN holds?
i.i.d.	Yes	All equal ($\mathbb{E}[X_i] = \mu$)	Yes
Markov chain (stationary)	No	All equal ($\mathbb{E}[X_i] = \mu$)	Yes
Markov chain (ergodic)	No	Different initially, converge to μ	Yes

5.3 Stationary and Ergodic Chains

A Markov chain is said to be **stationary** if its distribution does not change with time:

$$\mathbb{P}(X_{n+1} = j) = \mathbb{P}(X_n = j) \quad \forall n.$$

In this case,

$$\mathbb{E}[X_n] = \mathbb{E}[X_1] = \mu,$$

and the Law of Large Numbers holds even though the X_i are dependent:

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} \mu.$$

If the chain is not stationary but is **ergodic** (meaning it forgets its initial state and converges to a unique stationary distribution), then

$$\mathbb{E}[X_i] \rightarrow \mu \quad \text{as } i \rightarrow \infty,$$

and the sample average still satisfies

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} \mu.$$

5.4 Stationary Chain

A probability distribution π on the state space S is called a *stationary distribution* of a Markov chain with transition matrix P if, when the chain starts with initial distribution $\pi^{(0)} = \pi$, the distribution remains unchanged at all future times. That is,

$$\pi^{(n)} = \pi \quad \text{for all } n \geq 0.$$

Using the evolution equation $\pi^{(n+1)} = \pi^{(n)}P$, this condition is equivalent to the requirement

$$\pi = \pi P,$$

or, componentwise,

$$\pi(j) = \sum_{i \in S} \pi(i)P(i, j), \quad j \in S.$$

Thus a stationary distribution is one that, when used as the initial law of the Markov chain, remains invariant under the dynamics: the chain started in π stays in π for all time.

5.5 Ergodic Chains

Even if the initial distribution is non-stationary, an **ergodic** chain converges to a unique stationary distribution:

$$\pi^{(n)} = \pi^{(0)}P^n \rightarrow \pi.$$

Thus, the Law of Large Numbers still holds:

$$\frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{a.s.}} \mu.$$

5.6 Local Forgetting

The **Markov property** expresses local forgetting:

$$\mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, X_{n-1}, \dots) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

Once the current state is known, the past is irrelevant.

5.7 Global Forgetting (Ergodicity)

In many chains, as $n \rightarrow \infty$, the effect of the initial state vanishes:

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = s_j \mid X_0 = s_i) = \pi_j,$$

where $\pi = (\pi_1, \dots, \pi_k)$ is the *stationary distribution*, independent of the start.

5.8 Evolution Equation

The evolution of state probabilities is governed by:

$$\boldsymbol{\pi}^{(n+1)} = \boldsymbol{\pi}^{(n)} P, \quad \boldsymbol{\pi}^{(n)} = \boldsymbol{\pi}^{(0)} P^n.$$

Different initial distributions (e.g., all vowels, all consonants, or mixed) will all converge to the same stationary distribution in ergodic chains.

6 Example: Two-State Markov Chain - Vowel Constant

6.1 Setup

$$S = \{V, C\}, \quad P = \begin{bmatrix} p_{VV} & p_{VC} \\ p_{CV} & p_{CC} \end{bmatrix}, \quad p_{VC} = 1 - p_{VV}, \quad p_{CV} = 1 - p_{CC}.$$

The stationary distribution $\boldsymbol{\pi} = [\pi_V, \pi_C]$ satisfies:

$$\begin{cases} \pi_V = \pi_V p_{VV} + \pi_C p_{CV}, \\ \pi_C = \pi_V p_{VC} + \pi_C p_{CC}, \\ \boldsymbol{\pi} P = \boldsymbol{\pi}, \\ \pi_V + \pi_C = 1. \end{cases}$$

6.2 Three different Transition Probability representation

Consider a simple two-state system:

$$S = \{V, C\},$$

where V represents a vowel and C a consonant. The transitions are given by

$$\begin{aligned} \mathbb{P}(X_{n+1} = V \mid X_n = V) &= 0.8, & \mathbb{P}(X_{n+1} = C \mid X_n = V) &= 0.2, \\ \mathbb{P}(X_{n+1} = V \mid X_n = C) &= 0.4, & \mathbb{P}(X_{n+1} = C \mid X_n = C) &= 0.6. \end{aligned}$$

or represented in matrix form

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}.$$

or represented in transition diagramm

If the current state is a vowel ($X_n = V$), the probability of the next state being a vowel is always 0.8, no matter how we reached the current vowel. Whether the previous sequence was “CVCV” or “VVVV” does not change the probability of the next step. Thus, only the *current state* matters for predicting the next one.

6.3 Iteration

6.3.1 One

The probability of transitioning depends only on the current state—how we got there is

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix} \Rightarrow \boldsymbol{\pi} = [0.667, 0.333].$$

Thus, vowels occur about 2/3 of the time, consonants 1/3, regardless of the starting state.

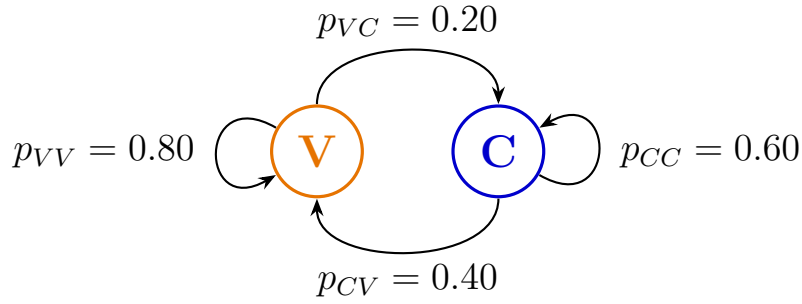


Abbildung 1.1: Vowel-Consonant Transition Diagram

6.3.2 Converging

The initial vector $\pi^{(0)}$ represents the probabilities of starting in each state. It can be any valid probability distribution (summing to 1), for example:

Starting condition	Initial vector $\pi^{(0)}$	Meaning
All vowels	$[1, 0]$	Certain start in V
All consonants	$[0, 1]$	Certain start in C
Half vowels, half consonants	$[0.5, 0.5]$	Mixed random start
30% vowels, 70% consonants	$[0.3, 0.7]$	Biased random start

Each of these starting distributions evolves over time according to

$$\pi^{(n)} = \pi^{(0)} P^n.$$

For example:

$$\text{If } \pi^{(0)} = [1, 0], \quad \pi^{(1)} = [0.8, 0.2], \quad \pi^{(2)} = [0.72, 0.28],$$

$$\text{If } \pi^{(0)} = [0, 1], \quad \pi^{(1)} = [0.4, 0.6], \quad \pi^{(2)} = [0.56, 0.44].$$

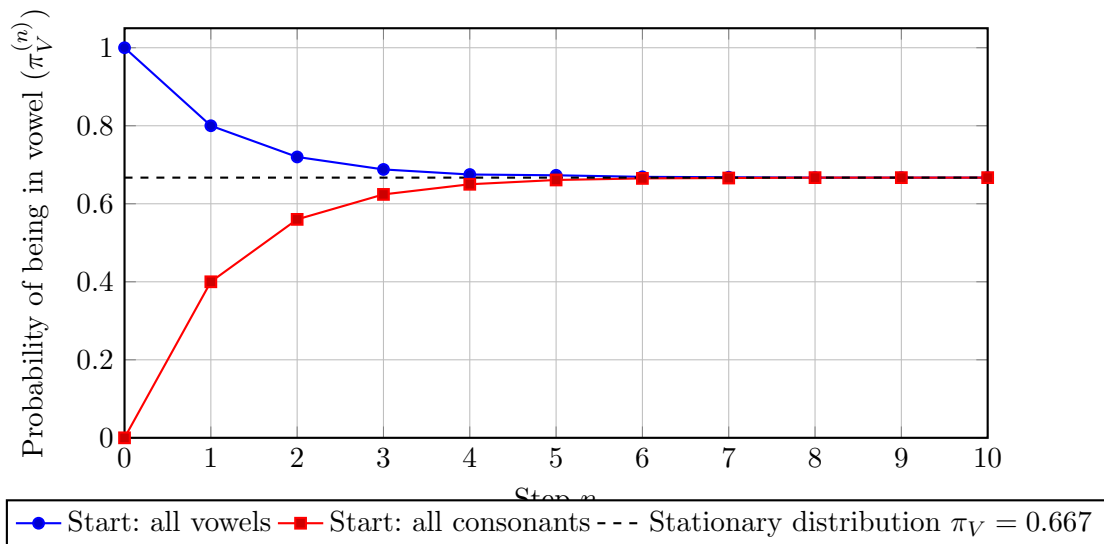
Although they start differently, both sequences converge to the same long-run equilibrium.

6.3.3 Visualization of Convergence (“Forgetting”)

For the matrix above,

$$\pi = [0.667, 0.333].$$

That means that, in the long run, the process will be in vowels about 66.7% of the time and in consonants about 33.3% of the time.



6.4 Interpretation

This plot shows how different initial starting conditions lead to the same long-run behavior. Even though one chain begins entirely in vowels and the other entirely in consonants, both converge to the same stationary distribution. This illustrates the *forgetting property* of Markov chains: the system “forgets” its initial state and stabilizes in equilibrium.

Teil II

Anhang

Anhang A

Abkürzungsverzeichnis

Anhang B

Glossar